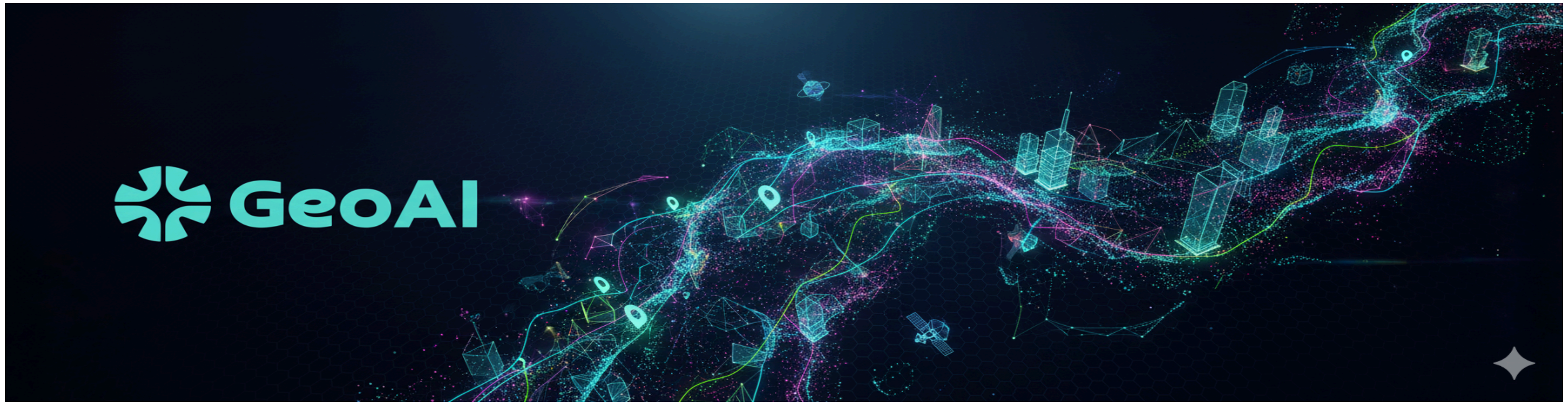
GeoAI

That's a wrap 🎬

Geospatial Graph Learning — the closing session

~30 min · delivered once, after Unit 5



TheArena
Upskilling Your Knowledge

One more pulse check — the *after*

The same anonymous survey we took on **day one** — shorter, and re-asked, so we can see how far this room actually moved. Scan now; it fills while we recap.

- **Seven** of the day-one questions, word-for-word: how you **use AI, write code, and check its work** — plus the graph/geo/transit fluency scales.
- Three new ones: can you now **DIRECT · CHOOSE · INTERPRET** — in your own read?
- And three **in your own words**: what you learned, whether the method worked, what you'll take with you. *(A sentence is plenty. We'll read them together — anonymously.)*

Scan to answer

Live counter on the screen →

Still no wrong answers — this is a **snapshot of the room**, ~3 minutes. Last one, promise.

GeoAI · Unit 6 — Course Closing

Movement 1 · The arc

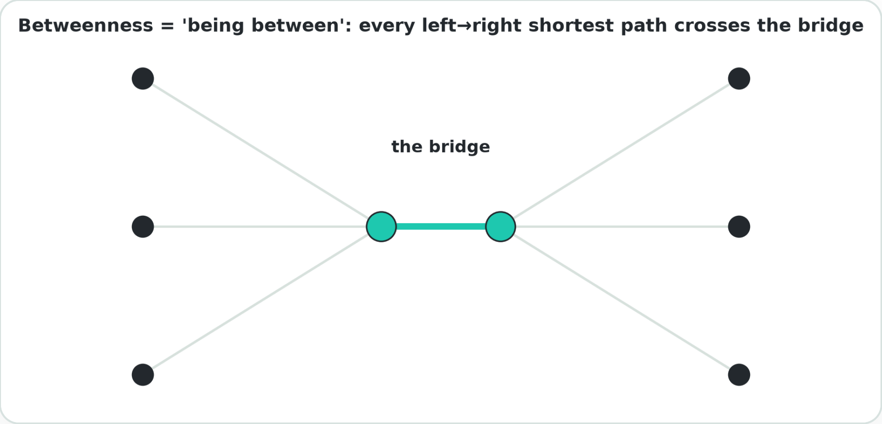
Five units, one skill per unit — in past tense

Unit 1 — Graph Substrate

The question we asked: *looking only at the road map, which streets are the arteries?*

Now you can…

- Turn any city's OSM extract into a **graph** — and defend every modeling choice (primal vs. dual, simplify or not, which CRS).
- Read **topology as signal**: betweenness finds the arteries before a single car moves.
- Say it in vocabulary an AI acts on: "*betweenness on the projected primal graph.*"



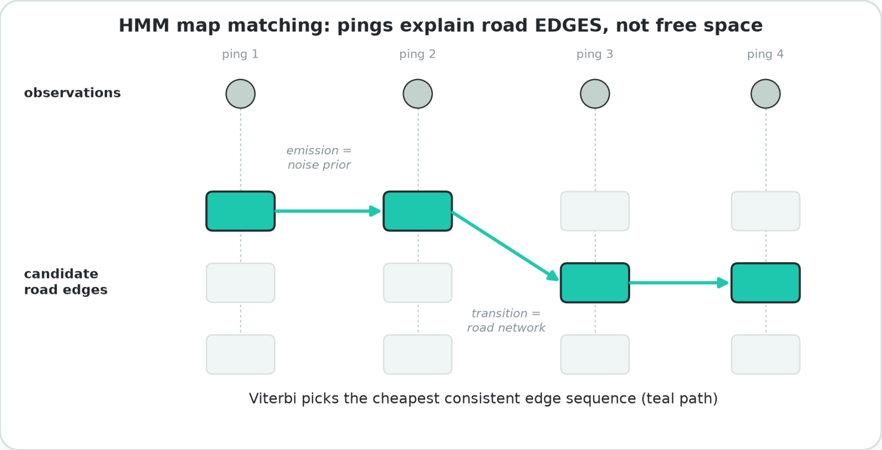
There is no neutral graph of a city — every modeling choice encodes an assumption.

Unit 2 — Trajectory Mining

The question we asked: *one bus pings once a minute — can you recover the route?*
Can many buses recover the whole network?

Now you can…

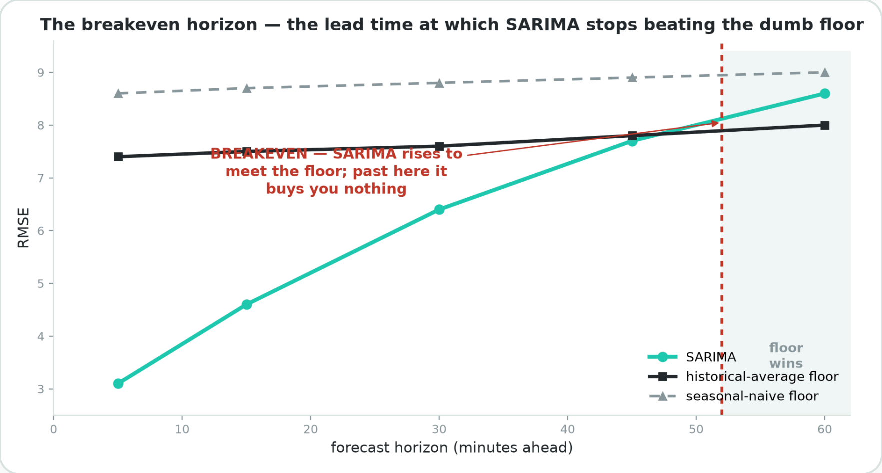
- Put noisy GPS **onto** the graph (HMM map matching, Viterbi) — and judge when to trust the match.
- Run the inference **backwards**: many traces → the network itself (map inference).
- Compare trajectories with the *right* distance — DTW vs. Fréchet, chosen by question.



Noise is a modeling decision, not a cleanup step.

Unit 3 — Statistical Baselines

The question we asked: *forecasting one sensor from its own past — how far ahead before a dumb baseline does just as well?*



Now you can…

- Fit and read ARIMA/SARIMA — stationarity, ACF/PACF, seasonality — without flinching.
- Find the **breakeven horizon**: where the fancy model stops earning its keep.
- Measure the **spatial gap** — the error that *time alone can't fix* because it can't see upstream.

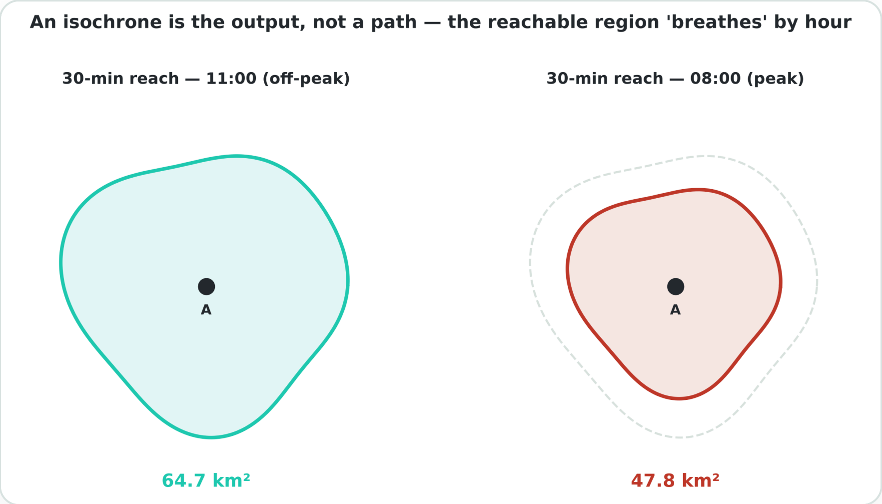
A baseline is the spec for "better."

Unit 4 — Dynamic Navigation

The question we asked: *car says 35 min, bus says 28 — which is THE answer, and what did each algorithm ignore?*

Now you can…

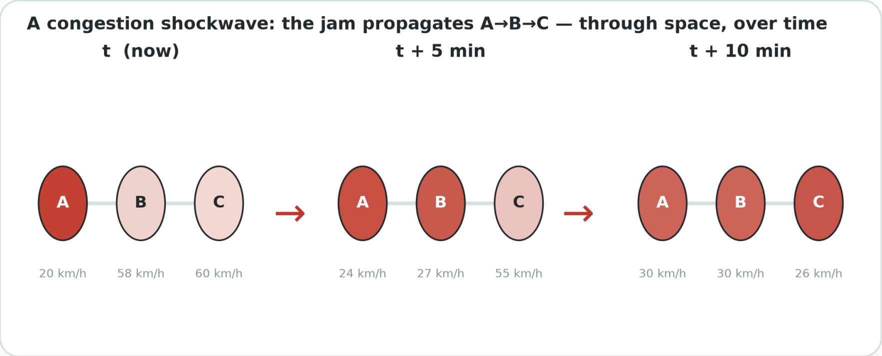
- Route under **time-varying cost** w(t) — and say when time-of-day actually matters.
- Pick the abstraction first, algorithm second: road graph → Dijkstra/A*/TDSP; schedules → RAPTOR/CSA.
- Read the **decision space**, not one path: isochrones, alternatives, the multimodal "Cost of Anarchy."



> **Choosing the cost IS choosing the question.**

Unit 5 — Spatio-Temporal GNNs

The question we asked: *can a model predict the congestion shockwave before it arrives — and can you read WHY from the model?*



Now you can…

- Build an ST-GNN (T-GCN/A3T-GCN) whose message passing closes the **gap U3 measured**.
- Demand that it **earn its complexity** — beat the baseline or don't ship it.
- Interrogate attention *carefully*: temporal attention is not spatial causality.

A model's attention is a hypothesis — readable, testable, and worth interrogating.

Five units, one pipeline

#	Unit	What it added to the shared graph
1	**Graph Substrate**	The graph itself — the object everything else lives on.
2	**Trajectory Mining**	Noisy GPS, matched *onto* that graph → routes & speeds.
3	**Statistical Baselines**	The forecast floor + the measured **spatial gap**.
4	**Dynamic Navigation**	Time-varying costs $w(t)$ on the graph → routing that reflects reality.
5	**Spatio-Temporal GNNs**	A learned model that closes U3's gap — and earns it.

One thread ran through all of it: a city **bus archive** — sparse GPS that U2 turned into routes, U3/U4 into speed forecasts and routing, U5 into a *learned* predictor. Five units; one dataset; one growing capability.

GeoAI · Unit 6 — Course Closing

Movement 2 · Working with an AI

What five units of supervised practice actually taught

The North Star — revisited

Verification is a consequence of doing CHOOSE + INTERPRET well — not the headline.

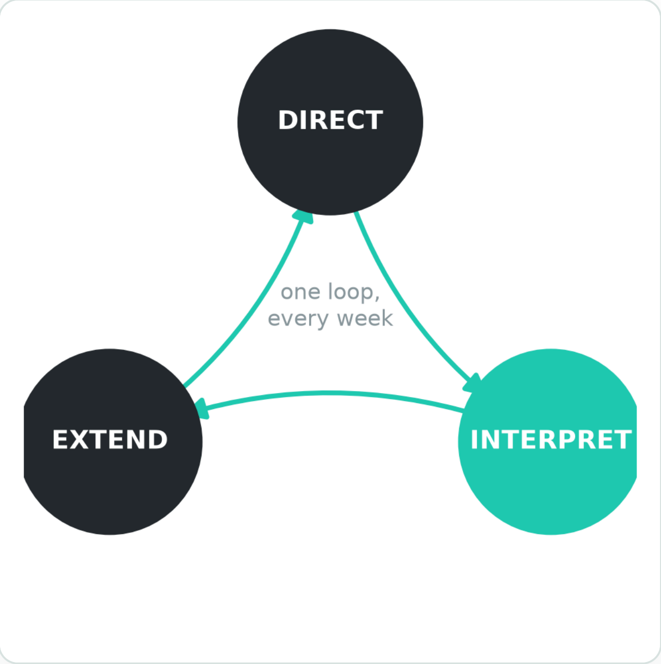
1. **DIRECT** an AI using **domain-precise vocabulary** — you did this every week.
2. **CHOOSE** the right analysis for a **real question** — including the wrong-class one.
3. **INTERPRET** results in **domain terms** — and chase the surprise, not bury it.

The loop you'll keep

Every supervised hour was the same loop — and it doesn't end with the course:

- **DIRECT** — phrase the request in precise, domain vocabulary.
- **INTERPRET** — restate the result in your own words; flag the surprises.
- **EXTEND** — ask the follow-up the result just earned.

The tools will change. **This loop won't.** It's how you stay the one supervising.



Take it with you: "*Restate what this result implies in domain terms, then propose the single most useful follow-up.*" — works on any problem, in any tool, next Monday.

Don't forget the *wrong-class* question

Every practice hid one question this unit's tools should **not** answer — and spotting it was half the skill.

- Centrality tells you the **arteries**, not the **traffic**.
- A forecast has a **breakeven horizon** — past it, the fancy model stops earning its keep.
- A GNN that doesn't beat the baseline is a GNN you **shouldn't ship**.

> The most valuable thing an expert does with an AI is **say no** to the analysis that doesn't fit — *before* running it, not after.

Two skills we never graded — keep practicing

Prompt engineering

Precise **DIRECT**ing *is* prompting: give context, name constraints, ask for the **reasoning**, show the shape you want. You got better at this without a rubric.

Working in loops

The first answer is rarely *the* answer. **Run → read → refine → run again.** You stopped one-shotting somewhere around Unit 2. Keep it.

> Ungraded, but they were the difference between a frustrating hour and a productive one — all term, and every day after.

Build your own AI project — like the one you just used

This course **is** an AI project: a repo the agent works *in*, not a chat it answers *in*.
Your fork already has the starter kit. Three setup habits that made it work:

1 · Give the agent a home.
A repo + one **context file** (your fork's `CLAUDE.md`): the goal, the rules, where things live, what "done" means. Every session starts already-briefed.

2 · Write your judgment down once.
`rubric.md`, the decision-log template — criteria the agent (and you) re-read every session instead of you re-typing them.

3 · Give repeatable jobs a role.
Not one mega-prompt: named **skills / subagents** with narrow briefs — a *researcher*, a *builder*, a *reviewer*. This course was literally built that way (researcher → demo-builder → practice-designer → slide-writer → a simulated *student* who dogfoods).

> Setup is a one-time cost. After that, **every session compounds** instead of starting over.

…and let the project evolve

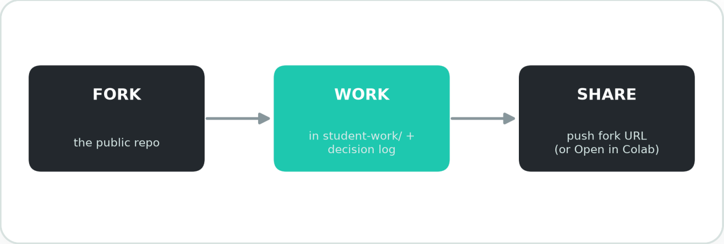
The setup isn't the trick — the **evolution** is. Habits that kept this course's project (and will keep yours) getting smarter:

- **Work in stages, with gates.** Research → design → build → verify — and a human *approves the outline before the build*. Cheap to redirect early, expensive late.
- **Keep resumable state.** A `STATE.md` ("where we are") + `NEXT.md` ("what's next") per workstream — any session, any agent, any *future you* picks up cleanly.
- **Verify before you ship.** Smoke tests, a fresh-eyes dogfood pass, Run-All on a clean runtime. The agent drafts; **evidence** decides.
- **Let the rules evolve too.** When you correct the agent twice, write it into the context file — the project learns, not just the chat.

> Start small: one repo, one context file, one rubric. Evolve the rest **when the project asks for it** — that's how this one grew.

Where to go next

- **Keep your fork.** The notebooks, decks, and `further-reading.md` per unit are yours — the sources are written so *your* agent can use them as context.
- **The workflow stays:** fork → work in `student-work/` → share. It's how you'll tackle the next problem that isn't in any syllabus.
- **`SYLLABUS.md`** remains your map — and the citations (`references.bib`) travel with you.



> The course ends; the **capability** doesn't. You now have the vocabulary to direct, the judgment to choose, and the eye to interpret — on graphs, geography, and movement.

GeoAI · Unit 6 — Course Closing

Movement 3 · The mirror

The room's own data — and its own words

Before → After — how far the room moved

Now the reveal: your **day-one answers** beside **today's**.

- The **operator headline** — relationship with AI, how code gets written, whether you *check* the output, end-to-end confidence.
- The **fluency delta** — graphs · geography · transit, day one → now.

Read the biggest mover aloud. Then find the one that barely moved — and ask why.

Live, on the room's own data

(Presenter, live) — `survey_explore.ipynb` → **Run all** → 🎯 **Before** → **After**.

A `direct → interpret → extend` demo, one last time — on ourselves.

In your own words

The three open questions, read the only way that's fair — **together, anonymously**:

- The **word cloud**: the room's own vocabulary. *(Did "graph" beat "prompt"? Did "baseline" make the cut?)*
- The **themes**: an LLM read every answer and extracted what you collectively said — with honest counts and verbatim quotes.
- The 🎯 **worth-chasing line**: the one surprising answer the analysis flagged.

Live, one more time

(Presenter, live) — `open_text_analysis.ipynb` → **Run all.**

*We just **directed** an AI at your words, we're **interpreting** its themes against the cloud, and the surprise is the **extend**.*

Thank you 🙏

You came in typing everything yourself.

You're leaving as someone who can direct, choose, and interpret — and who reads the code when it smells.

The robots are lucky to have you supervising.

GeoAI

Rights & disclaimer

Built with AI assistance. These slides, notebooks, and notes were drafted with AI assistance (Claude Code) and **reviewed by a human instructor** — but they **may still contain mistakes**. **Verify before you rely on anything here.**

> The closing survey is anonymous — we compare aggregates and read themes, never individuals.